



Ducaltus | AI Assurance & Governance

Trusted evaluation, oversight, and deployment assurance for high-stakes AI systems.

Research-Led AI Assurance for Real-World Deployment

Ducaltus is a research-led AI assurance and governance initiative focused on evaluating whether AI systems are reliable, fair, governable, and suitable for deployment within high-impact environments.

Website
ducaltus.com

Email
hello@ducaltus.com

LinkedIn
linkedin.com/company/ducaltus

About Ducaltus

Ducaltus is focused on advancing AI assurance methodologies for high-stakes artificial intelligence systems operating within sensitive, operational, or decision-critical environments.

As AI systems increasingly influence real-world decisions, traditional evaluation approaches based solely on aggregate accuracy may fail to identify hidden subgroup disparities, operational instability, and deployment risk.

Ducaltus supports organisations in evaluating whether AI systems are not only performant, but also reliable, governable, and suitable for deployment.

The organisation's work draws upon ongoing research in:

- Fairness metric disagreement
- Subgroup reliability analysis
- Intersectional fairness evaluation
- Deployment risk assessment
- Threshold and error trade-off analysis
- High-stakes AI assurance methodologies

Ducaltus operates at the intersection of technical evaluation, operational risk analysis, and responsible AI governance.

Why AI Assurance Matters

High overall model accuracy does not necessarily indicate reliable or responsible deployment behaviour.

AI systems operating in real-world environments may exhibit:

- Hidden subgroup performance disparities
- Inconsistent fairness outcomes across evaluation metrics
- Unequal false positive and false negative impacts
- Threshold-sensitive deployment instability
- Operational risks within high-impact decision environments

These risks become increasingly significant in domains where automated decisions may carry legal, ethical, operational, or public-impact consequences.

Core Capabilities

AI Risk Auditing

Structured assessment of model behaviour, deployment exposure, operational failure risk, and evaluation blind spots.

Fairness & Bias Evaluation

Analysis of subgroup disparities, fairness inconsistencies, and unequal error distributions across demographic and operational groups.

Deployment Assurance

Evaluation of whether AI systems are suitable for deployment within sensitive, high-impact, or decision-critical environments.

Subgroup Reliability Analysis

Assessment of model behaviour beyond aggregate accuracy to identify hidden performance failures affecting specific populations or operational conditions.

Threshold & Error Trade-off Assessment

Review of false positive and false negative trade-offs under varying thresholds, deployment settings, and operational constraints.

Governance & Oversight Support

Support for evaluation documentation, accountability processes, assurance reporting, and responsible AI governance practices.

Research & Methodology

Ducatus builds upon ongoing research exploring the relationship between fairness evaluation, deployment uncertainty, operational risk, and real-world AI system behaviour.

Fairness Metric Disagreement & FDI

Development of the Fairness Disagreement Index (FDI), designed to quantify disagreement between fairness metrics and evaluate the stability and reliability of fairness assessment itself.

Decision-Aware Deployment Risk

Research examining how disagreement between fairness measures may influence deployment uncertainty, operational reliability, and risk-sensitive decision environments.

Intersectional Fairness Evaluation

Methodologies focused on analysing subgroup behaviour across intersecting demographic characteristics rather than isolated protected attributes alone.

High-Stakes AI Assurance

Research-oriented approaches for evaluating whether AI systems are suitable for deployment in environments where failures may carry operational, legal, ethical, or societal consequences.

Selected Research

When Fairness Metrics Disagree

Research introducing the Fairness Disagreement Index (FDI) for analysing disagreement between fairness metrics and its implications for AI reliability assessment.

arXiv:

<https://arxiv.org/abs/2604.15038>

Reliability & Risk in AI-Assisted Medication Systems

Research exploring reliability, decision risk, and failure modes in AI-assisted systems operating in sensitive decision environments.

arXiv:

<https://arxiv.org/abs/2604.01449>

Fairness in Law Enforcement Facial Recognition

Research focused on fairness, deployment considerations, and subgroup performance analysis in high-impact facial recognition systems.

arXiv:

<https://arxiv.org/abs/2603.28675>

Current Research Direction

FairRisk-FDI, Intersectional Fairness & Deployment Risk

Current work focuses on extending fairness disagreement analysis into decision-aware deployment risk, intersectional subgroup evaluation, and practical assurance methodologies for high-stakes AI systems.

Founder

Khalid Adnan Alsayed

AI fairness and deployment risk researcher focused on:

- Fairness metric disagreement
 - Operational AI risk
 - Subgroup reliability analysis
 - Intersectional fairness evaluation
 - High-stakes AI assurance methodologies
-

Academic Background

- BSc (Hons) Artificial Intelligence
 - Teesside University
 - FdSc Software Engineering (Distinction)
-

Contact

Ducaltus

AI Assurance & Governance

Website	Email	LinkedIn
https://ducaltus.com	hello@ducaltus.com	https://www.linkedin.com/company/ducaltus/

© 2026 Ducaltus. All rights reserved.